# KUMARAGURU COLLEGE OF TECHNOLOGY

**Department of Artificial Intelligence and Data Science**

**DATA SCIENCE AND VISUALIZATION (24ADI204)**

WEEK 6 REPORT

Feature Engineering

Youth Smoking and Drug Consumption Analysis

**DATA-VIZ HACK-A-BIT 2026–2027**

**Team 16**

| Team Member Name | Roll Number |
|---|---|
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## 1. Week 6 Contribution

Member 3 – Implemented scaling, encoding and skewness correction.

## 2. Work Carried Out

During Week 6, I worked on feature engineering using the cleaned UCI Drug Consumption dataset in the project notebook. The work focused on preparing numerical and categorical features through encoding and scaling, checking skewness and grouping the Nscore feature into categories.

## 3. Feature Engineering Techniques

- Feature identification – Seven numerical personality features were used: Nscore, Escore, Oscore, Ascore, Cscore, Impulsive and SS. The 19 drug-consumption features contained categories from CL0 to CL6.
- Label encoding – The drug-consumption categories CL0 to CL6 were mapped to numeric values 0 to 6. The label-encoded dataset retained 1,885 rows and 32 columns.
- One-hot encoding – Gender was represented using two binary columns, Gender_Label_Female and Gender_Label_Male. The resulting dataset contained 1,885 rows and 33 columns.
- Standardization – StandardScaler was applied to the seven personality features. Their transformed means were approximately 0 and standard deviations approximately 1.
- Min-Max scaling – MinMaxScaler was applied to the same seven features, bringing their values into the range 0 to 1.
- Skewness analysis – All seven personality features had absolute skewness values below the notebook threshold of 0.5. Therefore, no feature was selected for Yeo-Johnson transformation; no unnecessary skewness correction was applied.
- Binning – Nscore was grouped into Low, Medium, High and Very High categories using equal-width and equal-frequency binning.

## 4. Visual Evidence from the Project Notebook

Figure 1: Skewness values of the seven personality features.

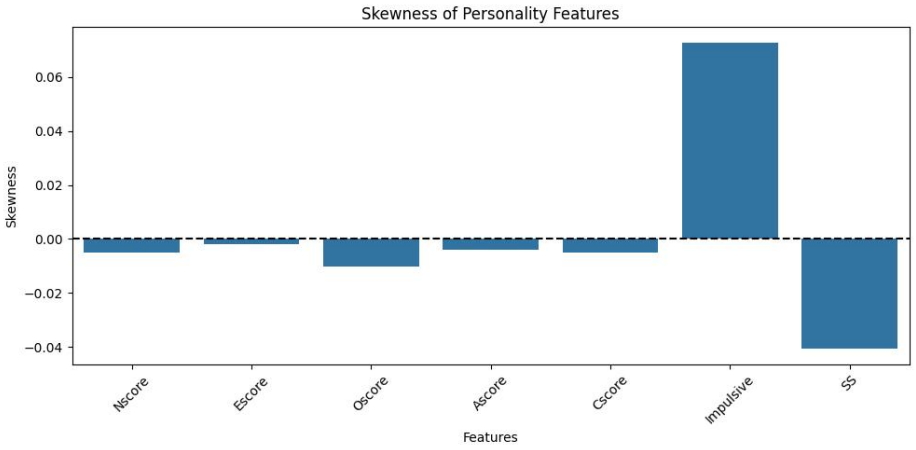



Figure 2: Skewness comparison before and after the Yeo-Johnson step. No transformation was required because all values were below the selected threshold.

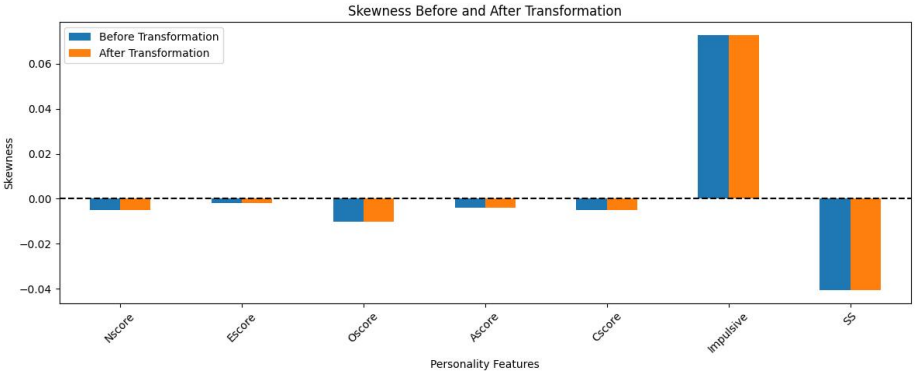
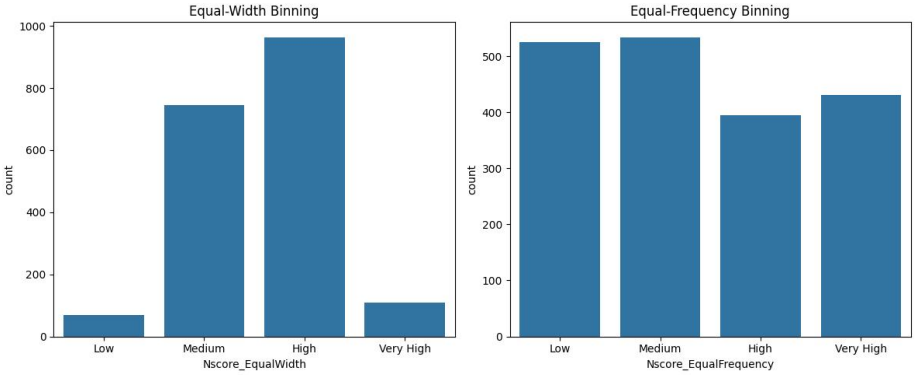



Figure 3: Comparison of equal-width and equal-frequency binning of Nscore.




## 5. Outcome of My Contribution

The notebook produced encoded and scaled versions of the cleaned dataset. The original dataset contained 1,885 rows and 32 columns; one-hot encoding increased the number of columns to 33, while the binned dataset contained 34 columns. Skewness analysis showed that the seven personality features were already close to symmetrical, so Yeo-Johnson transformation was not needed. These steps prepared the features for further analysis.